

Exercise 10

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Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using javascript

AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the chart and include Chart.js.

html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0"> <title>Inventory Management Visualization</title>
<style>
  body {
    font-family: Arial, sans-serif;
    text-align: center;
    margin: 50px;
  }
  canvas {
    margin: 20px auto;
```

```

    }
  </style>
</head>
<body>
  <h1>Inventory Management System</h1>
  <canvas id="pieChart" width="400"
  height="400"></canvas> <canvas id="barChart"
  width="400" height="400"></canvas> <script
  src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <script src="script.js"></script>
</body>
</html>

```

❏ Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic. javascript

```

// Data for the inventory
const inventoryData = {
  labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
  datasets: [
    {
      label: 'Items in Stock',
      data: [200, 150, 100, 80, 50],
      backgroundColor: [
        '#FF6384',
        '#36A2EB',

```

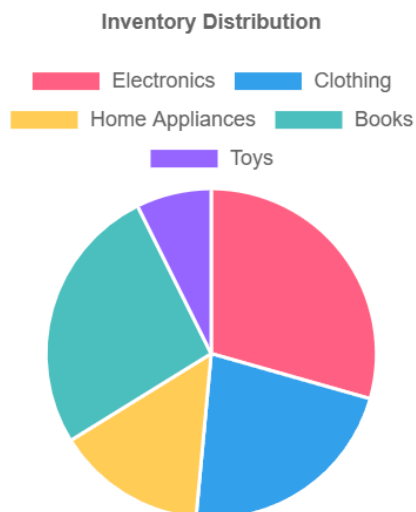
```

        '#FFCE56',
        '#4BC0C0',
        '#9966FF'
    ],
}
]
};

```

Inventory Management System

Pie Chart



// Creating the Pie Chart

```

const ctxPie = document.getElementById('pieChart').getContext('2d');
const pieChart = new Chart(ctxPie, {
  type: 'pie',
  data: inventoryData,
  options: {
    responsive: true,
    title: {
      display: true,

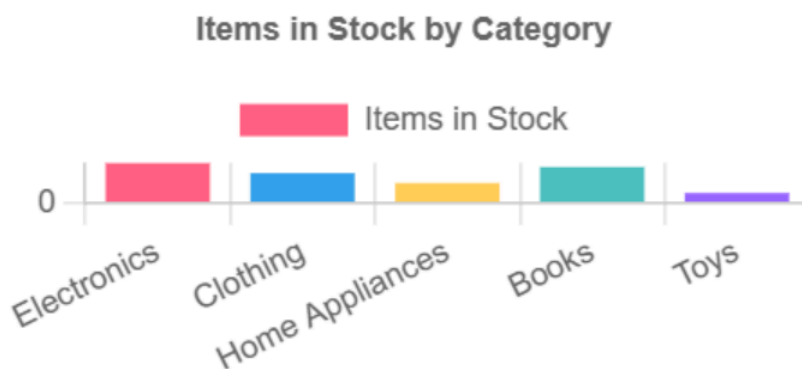
```

```

        text: 'Inventory Distribution' } }
    }); // Creating the Bar Chart
const ctxBar = document.getElementById('barChart').getContext('2d');
const barChart = new Chart(ctxBar, {
    type: 'bar',
    data: inventoryData,
    options: {
        responsive: true,
        title: {
            display: true,
            text: 'Items in Stock by
Category' },
        scales: {
            yAxes: [{
                ticks: {
                    beginAtZero: true } } ] } } });

```

Bar Chart



Result: The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript was created.

